1 Chiziqli tenglamalar va tengsizliklar

1-to'plam. *Ko'p qadamli tenglamalar (Book 2 darajasida). x ni toping.*

1. $\frac{1}{2}(x+4) = x-1$

2. $3(x-2) = 2(x+1)$

3. $\frac{2}{3}x + 1 = \frac{1}{3}x + 5$

4. $5 - 2(x-3) = x+1$

5. $\frac{3}{4}(x-4) = 2x-7$

6. $4(x+1) - 3 = 2(x+5)$

2-to'plam. *"Nima so'ralyapti" qopqoni – tenglamani yeching, lekin aynan so'ralgan qiymatni bering, shunchaki x ni emas.*

1. Agar $3x+5 = 20$ bo'lsa, $6x+10$ nechiga teng?

2. Agar $2(x-3) = 10$ bo'lsa, $x-3$ nechiga teng?

3. Agar $4x - 7 = 9$ bo'lsa, $2x$ nechiga teng?

4. Agar $\frac{x+2}{3} = 5$ bo'lsa, $x+2$ nechiga teng?

5. Agar $5x+1 = 26$ bo'lsa, $10x+2$ nechiga teng?

6. Agar $3(2x+1) = 21$ bo'lsa, $2x+1$ nechiga teng?

3-to'plam. *Tengsizliklar – ishora almashinuviga e'tibor bering. x ni toping.*

1. $2x + 3 < 11$

2. $-4x + 1 \geq 13$

3. $5 - x \leq 8$

4. $\frac{1}{2}x - 3 > 1$

5. $-3(x - 2) < 9$

6. $7 - 3x \geq -5$

4-to'plam. *Qo'shma tengsizliklar. x ni toping.*

1. $-2 < x + 3 \leq 7$

2. $3x - 1 < -4$ yoki $3x - 1 > 8$

3. $-5 \leq 2x + 1 < 9$

4. $x - 4 < -6$ yoki $x - 4 > 2$

5. $-1 < \frac{x + 1}{2} \leq 3$

6. $2x + 3 < 1$ yoki $2x + 3 > 9$

5-to'plam. *So'zlardan tengsizlikka o'tkazing, so'ng yeching.*

1 Biror son 5 ga oshirilsa, natija kamida 12 bo'ladi. (Sonni toping.)

2 Biror sonning uch barobari 21 dan oshmaydi.

3 Biror son 4 ga kamaytirilsa, natija 10 dan kichik.

4 Biror sonning ikki barobari, 3 ga kamaytirilganda, kamida 7.

5 Biror son kamida -1 va ko'pi bilan 5. (Faqat qo'shma tengsizlikni yozing – yechish shart emas.)

6 Sonning ikki barobaridan besh kam – 15 dan oshmaydi. Sonning eng katta butun qiymati nechiga teng?

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1 $-2x > 8$ ifodasida ko'rsatilgan yechimda xato bor:

"Ikkala tomonni -2 ga bo'ling: $x > -4$."

Xato va to'g'ri javob: .....

2 $x-5 > 2$ ifodasida ko'rsatilgan yechimda xato bor:

"Manfiy son bor, shuning uchun ishora almashadi: $x < 7$."

Xato va to'g'ri javob: .....

3 Agar $4x - 1 = 15$ bo'lsa, $4x$ nechiga teng ifodasida xato bor:

"$x = 4$, demak javob 4."

Xato va to'g'ri javob: .....

4 $-2(x - 3) < 10$ ifodasida ko'rsatilgan yechimda xato bor:

"Taqsimlang: $-2x - 6 < 10$, demak $x > -8$."

Xato va to'g'ri javob: .....

5 $-4 < 2x < 8$ ifodasida ko'rsatilgan yechimda xato bor:

"O'rta va o'ng tomonni 2 ga bo'ling: $-4 < x < 4$."

Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin – eng qiyin topshiriqlar.*

1 Agar $\frac{2}{3}(x-3) = x-5$ bo'lsa, x ni, so'ng $3x$ ni toping.

2 $-5 < 3x+1 \leq 10$ ni yeching. Qo'shma tengsizlikni yozing, so'ng shartni qanoatlantiruvchi eng kichik butun x qiymatini toping.

3 Yetti minus sonning ikki barobari kamida -3. Sonning eng katta qiymati nechiga teng?

4 $4(x-1)-2x = 3(x+2)-10$ ni x uchun yeching, so'ng $x+2$ ni toping.

5 $-6 \leq -2(x-1) < 4$ ni yeching.

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	$x = 6$	2	$x = 8$
	3	$x = 12$	4	$x = \frac{10}{3}$
	5	$x = \frac{16}{5}$	6	$x = \frac{9}{2}$
2-to'plam	1	40	2	5
	3	8	4	15
	5	52	6	7
3-to'plam	1	$x < 4$	2	$x \leq -3$
	3	$x \geq -3$	4	$x > 8$
	5	$x > -1$	6	$x \leq 4$
4-to'plam	1	$-5 < x \leq 4$	2	$x < -1$ yoki $x > 3$
	3	$-3 \leq x < 4$	4	$x < -2$ yoki $x > 6$
	5	$-3 < x \leq 5$	6	$x < -1$ yoki $x > 3$
5-to'plam	1. $x + 5 \geq 12 \Rightarrow x \geq 7$			
	2. $3x \leq 21 \Rightarrow x \leq 7$			
	3. $x - 4 < 10 \Rightarrow x < 14$			
	4. $2x - 3 \geq 7 \Rightarrow x \geq 5$			
	5. $-1 \leq x \leq 5$			
	6. $2x - 5 \leq 15 \Rightarrow x \leq 10$, eng katta butun $= 10$			
6-to'plam	1. Xato – manfiyga bo'lganda ishora almashishi kerak, lekin yo'nalish teskari ko'rsatilgan. To'g'ri: $x < -4$.			
	2. Xato – manfiyga ko'paytirish/bo'lish bo'lmagan (faqat ayirish), ishora almashmasligi kerak. To'g'ri: $x > 7$.			
	3. Xato – $4x$ o'rniga x berilgan. To'g'ri: $x = 4$, demak $4x = 16$.			
	4. Xato – $-2 \times (-3)$ ni -6 deb taqsimlagan, $+6$ bo'lishi kerak edi. To'g'ri: $-2x + 6 < 10 \Rightarrow x > -2$.			
	5. Xato – o'rta va o'ng tomon 2 ga bo'lingan, lekin -4 bo'linmay qolgan. To'g'ri: $-2 < x < 4$.			
7-to'plam	1. $x = 9$, $3x = 27$			
	2. $-2 < x \leq 3$, eng kichik butun $= -1$			
	3. $x \leq 5$, eng katta qiymat $= 5$			
	4. $x = 0$, $x + 2 = 2$			
	5. $-1 < x \leq 4$			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

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